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Problem

ONE-STEP EDITING COLLAPSES

- Fast generators such as SD-Turbo synthesize in one step, yet conventional training-free editors depend on many small updates.
- Naively subtracting target and source drifts yields a high-energy, nonlinear and noisy control field.
- A single large integration step then distorts the edited object and destroys unedited backgrounds.

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Transport view

EDITING AS LOW-ENERGY FLOW

- Treat source- and target-prompt image distributions as endpoints of a dynamic optimal-transport problem.
- The ideal field minimizes kinetic energy subject to the continuity equation, but only noisy model queries are observable.
- Source/target outputs are mapped into a shared drift domain, supporting noise-, velocity-, score- and x0-parameterized models.

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Chord control

TEMPORAL SMOOTHING

- Average the observable residual across a short causal time window instead of using the instantaneous difference.
- $u_{\text{hat}} = [t R(t-\text{delta}) + \text{delta} R(t)] / (t + \text{delta})$
- The positive kernel is an L2 energy contraction, suppressing spikes and creating a stable chord for one-step integration.

CHORDEDIT

ChordEdit

One-Step Low-Energy Transport for Image Editing | 2026

Replace the unstable difference between source- and target-conditioned one-step fields with a temporally smoothed, optimal-transport-inspired control chord that preserves structure in one large step.

one-step editing

optimal transport

training-free

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Algorithm

TRANSPORT, THEN REFINE

- Query the model on a synthetic noisy state, compute the chord field and update $x_{\text{pred}} = x_{\text{src}} + \text{lambda } u_{\text{hat}}$.
- The transport uses one NFE, is model-agnostic, training-free and inversion-free, and works with a single noise sample.
- An optional one-NFE proximal prediction step strengthens target semantics while keeping transport separate from refinement.

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Evidence

REAL-TIME WITH FIDELITY

- PIE-bench transport-only editing reaches PSNR 23.89 at 0.20 s; full ChordEdit reaches CLIP-Edited 22.96 at 0.38 s.
- It is 19x faster than FlowEdit, over 208x faster than Direct Inversion and uses under half the VRAM of SwiftEdit.
- Across SD-Turbo, SwiftBrush-v2 and InstaFlow it beats the naive baseline; one sample is seed-stable and Pareto-dominant.

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Meaning & risks

STABILITY ENABLES THE STEP

- The contribution is not merely fewer solver steps: smoothing changes the field so a coarse step becomes numerically viable.
- The chord prioritizes preservation; proximal refinement separately controls the semantic-strength trade-off.
- Real-time editing can be misused for deceptive manipulation, so the authors target creative and assistive applications.